

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyse DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)

Assessment Tool Weightage: 20%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember

Code of Conduct:

I Shaik Ayesha Tabassum(192525074) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

S.Ayesha Tabassum

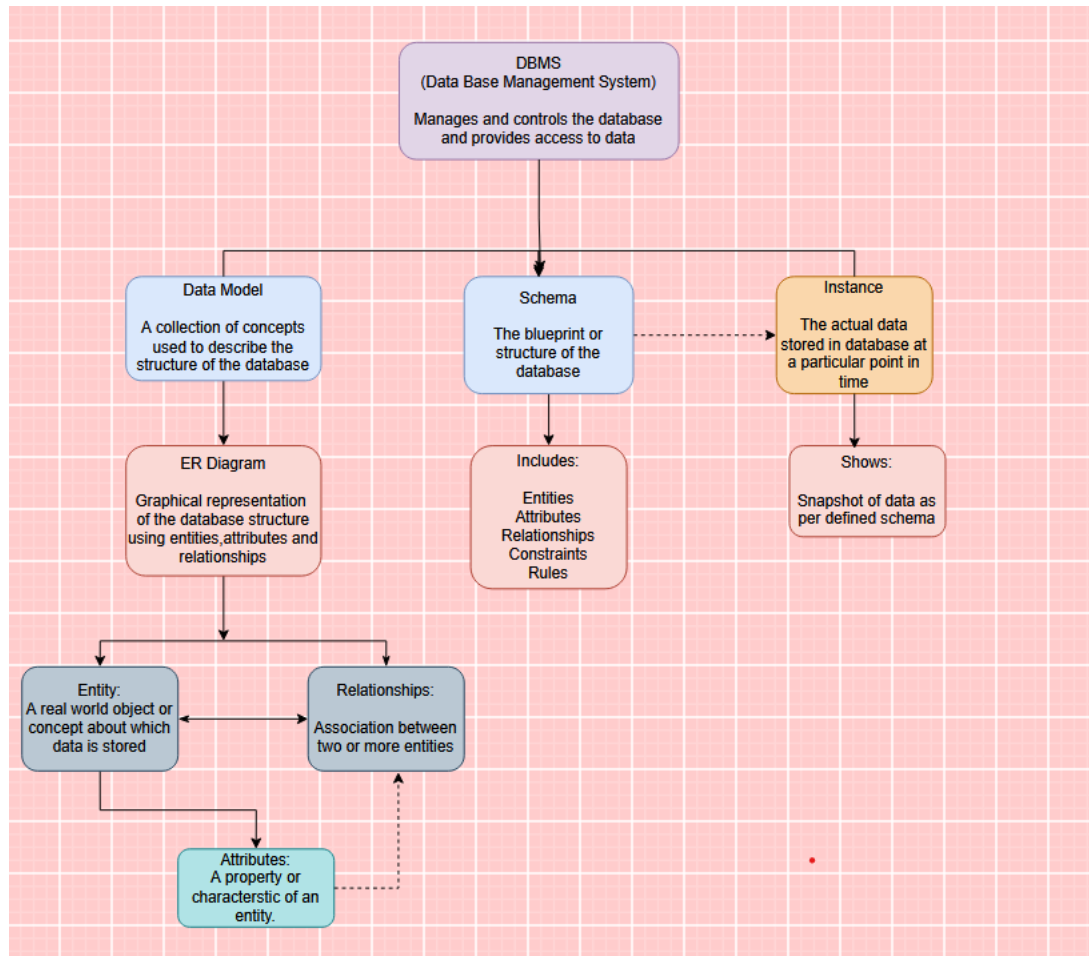
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.



DBMS (Database Management System)

A **Database Management System (DBMS)** is software that stores, manages, and retrieves data efficiently. It provides security, reduces data redundancy, and allows multiple users to access the database simultaneously.

Data Model

A **Data Model** defines how data is organized, stored, and related within a database. It provides the structure for designing databases.

Schema

A **Schema** is the logical blueprint of a database. It defines the tables, fields, relationships, and constraints, and usually remains unchanged unless the database structure is modified.

Instance

An **Instance** is the actual data stored in the database at a particular point in time. Unlike the schema, it changes whenever data is inserted, updated, or deleted.

ER Diagram (Entity-Relationship Diagram)

An **ER Diagram** is a graphical representation of a database. It illustrates entities, attributes, and relationships, helping designers visualize the database structure before implementation.

Entity

An **Entity** is any real-world object, person, place, or thing about which data is stored.

Examples: Student, Employee, Product, Customer.

Attribute

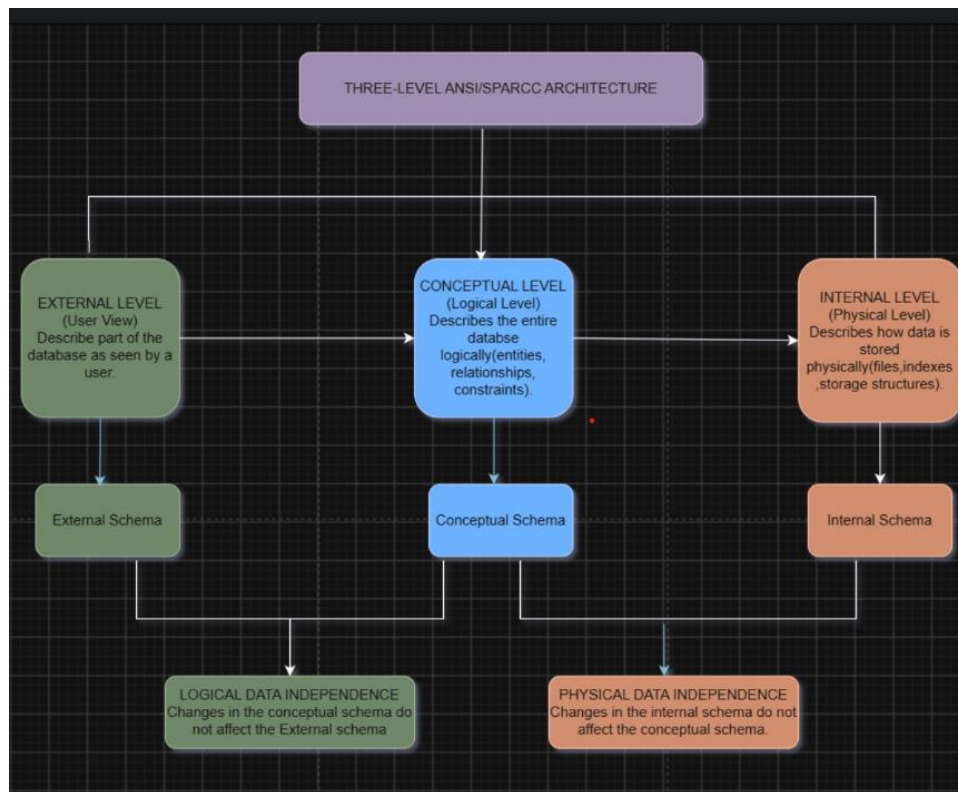
An **Attribute** is a property or characteristic of an entity. It provides detailed information about that entity.

Example: For a Student entity, attributes can be Student_ID, Name, Department, and Age.

Relationship

A **Relationship** describes how two or more entities are connected. It shows the association between entities in a database.

2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.



ANSI/SPARC Architecture

The ANSI/SPARC architecture is a **three-level database architecture** that separates user applications from physical database storage. It improves flexibility, security, and data independence.

1. External Level (View Level)

- Represents the user view of the database.
- Different users can access different views.
- Hides unnecessary data from users.

2. Conceptual Level (Logical Level)

- Represents the complete logical structure of the database.
- Defines tables, relationships, constraints, and data types.
- Independent of physical storage.

3. Internal Level (Physical Level)

- Represents how data is physically stored.
- Includes files, indexes, storage devices, and access methods.
- Focuses on storage efficiency and performance.

Data Independence

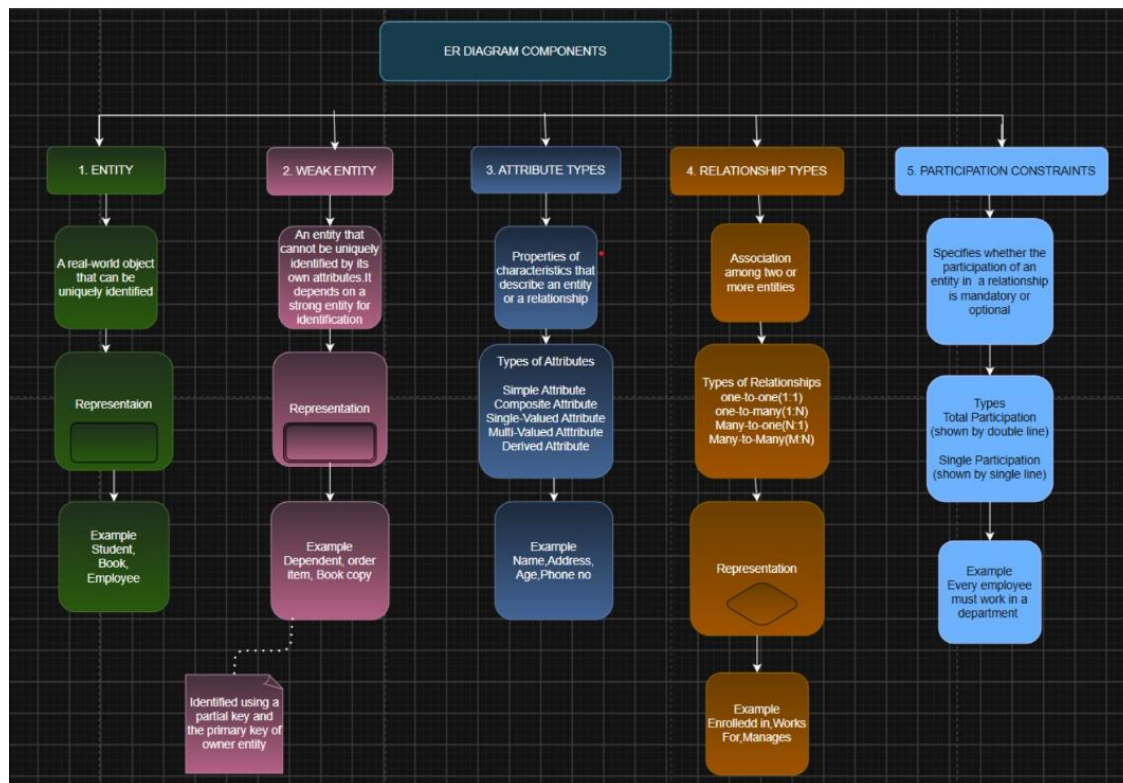
Logical Data Independence

- Ability to change the **Conceptual Level** without affecting the **External Level**.
- Example: Adding a new column to a table without changing user views.

Physical Data Independence

- Ability to change the **Internal Level** without affecting the **Conceptual Level**.
- Example: Changing file organization or adding indexes without changing the database schema.

4. Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.



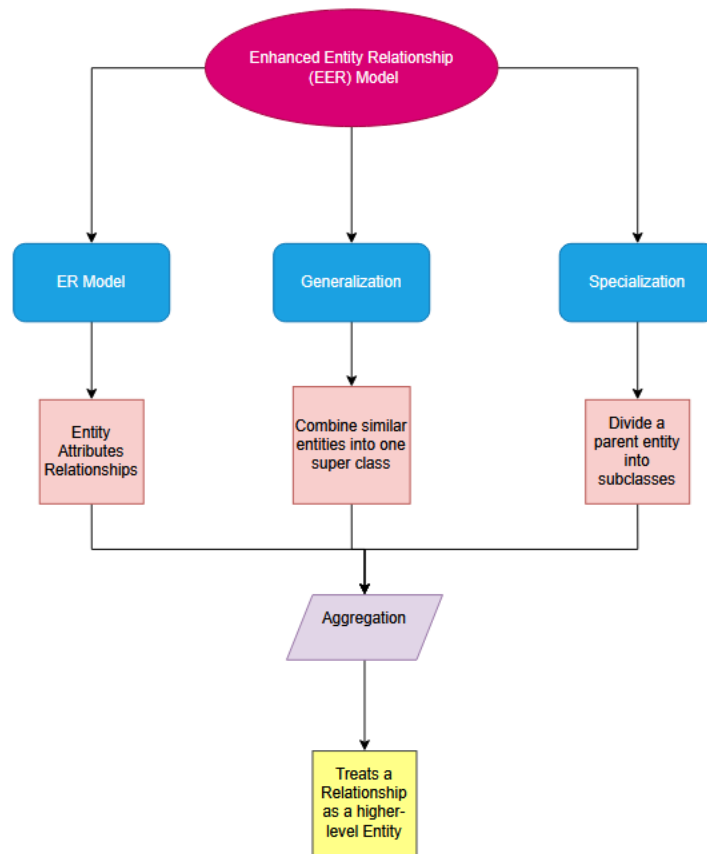
An **Entity-Relationship (ER) Diagram** is a graphical representation of a database that shows entities, attributes, relationships, and constraints. It helps in designing an efficient database structure.

- **Entity:** A real-world object or person about which data is stored (e.g., Student, Employee).
- **Weak Entity:** An entity that cannot exist without a related strong entity and depends on it for identification.
- **Attribute Types:** Describe the properties of an entity, such as **Simple**, **Composite**, **Single-Valued**, **Multi-Valued**, **Derived**, and **Key Attributes**.
- **Relationship Types:** Define how entities are connected, including **One-to-One (1:1)**, **One-to-Many (1:N)**, **Many-to-One (N:1)**, **Many-to-Many (M:N)**, and **Recursive Relationships**.
- **Participation Constraints:** Specify whether an entity's participation in a relationship is **Total** (mandatory) or **Partial** (optional).

Conclusion

ER diagram components help represent real-world data clearly and accurately, making database design simple, organized, and efficient.

6. Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.



The **Enhanced Entity-Relationship (EER) Model** is an extension of the basic ER model. It provides advanced features to represent complex real-world data more accurately.

- **Generalization:** Combines similar lower-level entities into a common higher-level **superclass**.
- **Specialization:** Divides a higher-level entity into smaller **subclasses** based on specific characteristics.
- **Aggregation:** Treats a relationship as a higher-level entity so it can participate in another relationship.

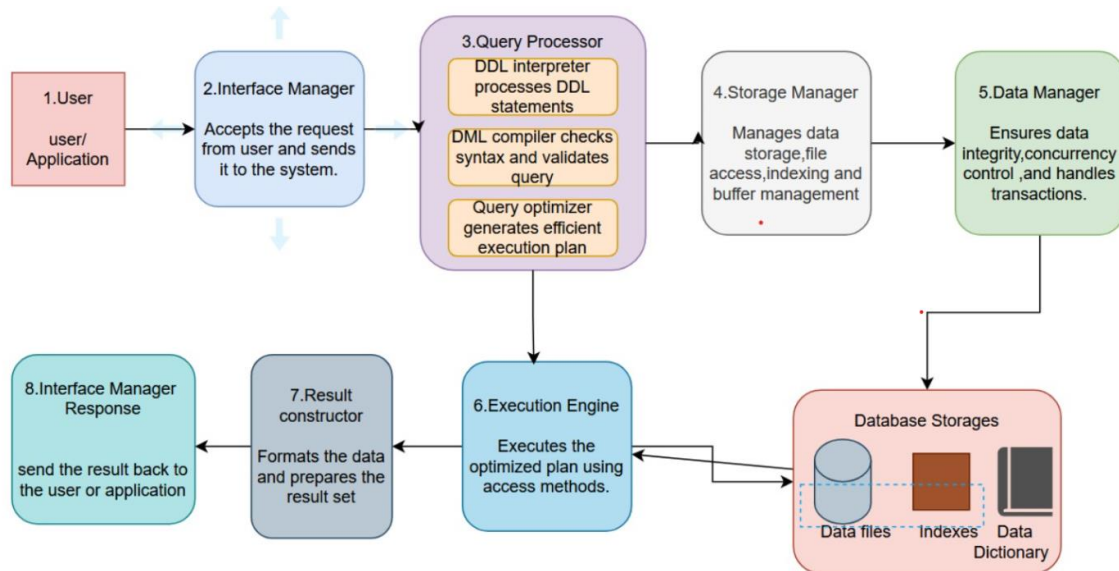
Key Points

- Extends the basic **ER Model**.
- Supports **inheritance** between entities.
- Represents complex relationships more effectively.
- Improves database design by reducing redundancy and increasing flexibility.

Conclusion

The **EER Model** enhances the ER model with **generalization, specialization, and aggregation**, making database design more powerful, flexible, and suitable for real-world applications.

7. Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.



DBMS Software Architecture – Short Matter

The **DBMS Software Architecture** shows how a user's request is processed and converted into the required data. It ensures efficient, secure, and reliable data retrieval.

- **User Request:** The user submits a query or request through an application.
- **Query Processor:** Analyzes, validates, and optimizes the query for execution.
- **Storage Manager:** Manages data files, indexes, and physical storage.
- **Database:** Stores the actual data and system catalog.
- **Execution Engine:** Retrieves the required data from the database.
- **Result:** The processed data is returned to the user.

Flow

User Request → Query Processor → Storage Manager → Database → Data Retrieval → Result to User

Conclusion

The DBMS software architecture efficiently processes user requests, retrieves accurate data, and ensures security, integrity, and fast database operations.